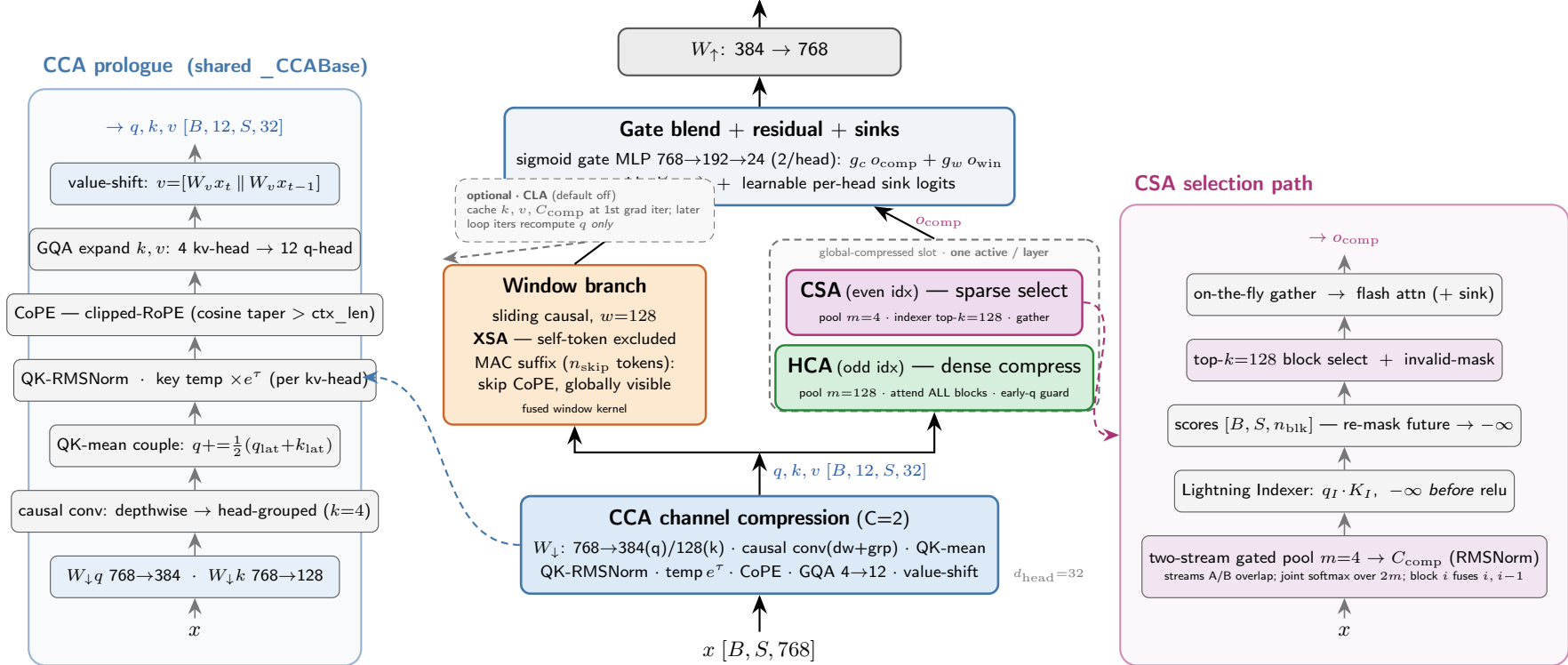


MORPH Attention — Triple-Axis Compression

CCA channel-compress feeds a local window || one global-compressed branch (CSA / HCA), blended by a per-head gate
attn out $[B, S, 768] \rightarrow$ block residual



Four compression axes (all faithful to morph/model/attention.py): channel CCA $E \rightarrow E/2$ select CSA top-128 of $S/4$ blocks dense HCA all $S/128$ blocks local window $w=128$.

CSA/HCA alternate on the *global* layer index (even $\rightarrow$ CSA, odd $\rightarrow$ HCA), baked at `__init__` — so a looped core layer keeps its variant on every iteration (*not* iteration-conditioned). Causal-leak fix: indexer masks future blocks to $-\infty$ *before* relu, and re-checks validity after gather.

Combine: per-head sigmoid gate gives 2 weights (compressed, window); residual- α adds the down-projected q_{lat} back; **attention sinks** are a learnable per-head logit appended to the compressed softmax (a no-op "null" key). Fused Triton kernels: CCA prologue, CCA conv, window, CSA, HCA (each fwd/grad-exact vs eager).

Notation: B batch, S seq, $H=12$ q-heads, $H_{kv}=4$ kv-heads (GQA $\times 3$), channel compression $C=2$, $d_{head}=768/(CH)=32$, $n_{blk}=S/m$. Color is secondary — every box is text-labeled.